

DATA AND INFORMATION QUALITY REPORT

December 2025

1 Project Report

Project ID: DaIQ-CRIMES-2025

Assigned Dataset: Boston Crime Incident Reports. The dataset consists of:

- `crime.csv`
- `offense_codes.csv`

Students:

- Daniele Spini – 279837
- Oksana Batonova – 274392
- Seyed Ali Chavoshian – 250888

2 Setup Choices

The project was developed in Python using the following libraries:

- `pandas`, `numpy` for data manipulation and numerical computation
- `matplotlib`, `seaborn`, `plotly` for exploratory and comparative visualization
- `ydata-profiling`, `sweetviz` for automated data profiling and data quality inspection
- `rapidfuzz`, `recordlinkage` for string similarity analysis and probabilistic record linkage
- `scikit-learn` for preprocessing, modeling, imputation, and evaluation

The workflow explicitly separates data profiling, data cleaning, and data analysis phases. This structure ensures transparency, reproducibility, and traceability of all data preparation decisions.

Data preparation adopts a **hybrid strategy** that combines rule-based validation with model-based techniques where deterministic corrections are insufficient. Rule-based methods were preferred whenever domain knowledge enabled unambiguous corrections. However, for complex missingness patterns and non-exact duplicates, statistical and machine-learning-based methods were employed to improve robustness and data completeness while preserving underlying data distributions.

Missing value handling leverages **MICE (Multiple Imputation by Chained Equations)** for numerical and spatial attributes and **KNN-based imputation** for categorical features. These techniques were selected to capture multivariate dependencies that cannot be reliably addressed through simple univariate or rule-based imputations.

Duplicate detection follows a two-stage strategy. Exact duplicates are handled deterministically, while non-exact duplicates are resolved using **probabilistic record linkage** with a supervised classification model. This approach allows duplicate resolution under uncertainty and avoids overly aggressive ID-based deletions.

The same downstream analysis pipeline was applied to both the dirty and cleaned datasets to ensure a fair and controlled comparison, allowing the impact of data quality improvements to be isolated from modeling choices.

To quantify the impact of these data quality improvements, a controlled downstream analysis was performed on both the dirty and cleaned datasets, as described in the following section.

3 Pipeline Implementation

3.1 Data Profiling and Data Quality Assessment

This phase establishes a baseline understanding of the dataset prior to any cleaning. Its objective is to identify structural properties and data quality issues that may affect downstream processing and analysis.

Data profiling and quality assessment were performed programmatically, following the same logical order as the analysis notebook. The workflow begins with a general inspection of the dataset and progressively focuses on missing values, uniqueness patterns, descriptive statistics, and cross-field consistency across temporal, spatial, and categorical attributes.

This structured approach ensures that all subsequent data preparation decisions are grounded in observed data characteristics rather than assumptions.

3.1.1 General Information on Data

The initial inspection summarizes the dataset schema and provides a representative snapshot of the records. The crime dataset contains 319,073 rows and 17

columns, covering identifiers, offense information, administrative areas, temporal attributes, and spatial coordinates.

The dataset includes both numerical and categorical attributes. Numerical fields include `OFFENSE_CODE`, `YEAR`, `MONTH`, `HOURL`, `Lat`, and `Long`, while categorical or textual attributes include `INCIDENT_NUMBER`, `OFFENSE_CODE_GROUP`, `OFFENSE_DESCRIPTION`, `DISTRICT`, `REPORTING_AREA`, `SHOOTING`, `DAY_OF_WEEK`, `UCR_PART`, `STREET`, and `Location`.

A preview of the first records was used to verify correct parsing and to identify early indications of missing values or formatting inconsistencies. This step confirmed that the dataset structure is suitable for detailed profiling and quality assessment.

3.1.2 Data Quality Assessment

Data quality was evaluated using multiple complementary dimensions, following established data quality frameworks. Each dimension was operationalized through explicit quantitative metrics computed directly from the dataset.

Completeness Completeness measures the proportion of non-null values relative to the total number of cells. Missing values were primarily concentrated in `SHOOTING`, spatial coordinates (`Lat`, `Long`), `STREET`, and `DISTRICT`.

Given the semantic meaning of `SHOOTING`, missing values were interpreted as non-shooting incidents and recoded as ‘‘N’’. This choice substantially improved completeness while preserving the intended meaning of the attribute.

Accuracy Accuracy evaluates conformity with reference data, valid ranges, and domain constraints. Checks included validation of `OFFENSE_CODE` against the reference table, range checks for temporal variables, and categorical validation for `DAY_OF_WEEK` and `UCR_PART`.

Fuzzy string similarity methods were applied to assess semantic consistency between offense descriptions, offense groups, and reference names. Both character-based and token-based similarities were used, revealing systematic variations attributable to legacy naming conventions rather than data corruption.

Consistency Consistency assesses logical coherence across related variables. Constraints included agreement between temporal fields and occurrence timestamps, spatial consistency between encoded locations and coordinates, and valid categorical encoding of `SHOOTING`.

Most inconsistencies were driven by missing spatial information rather than conflicting values. No systematic temporal or categorical contradictions were detected.

Uniqueness Uniqueness was evaluated for `INCIDENT_NUMBER`. Although exact duplicate rows were rare, repeated identifiers were observed, reflecting legitimate multi-record representations of the same incident.

This finding motivated a conservative deduplication strategy in later stages, avoiding automatic removal based solely on identifiers.

Validity Validity checks verified that values conform to syntactic rules and admissible ranges. Offense codes matched the reference table, temporal values fell within expected bounds, and categorical attributes adhered to predefined value sets.

Spatial validity checks confirmed that latitude and longitude values were numeric and globally valid. Violations were rare and mainly associated with missing values.

Plausibility Plausibility evaluates alignment with real-world constraints. Geographic plausibility checks confirmed that spatial coordinates fall within the city boundaries. Temporal plausibility checks verified realistic distributions of years, months, and days of the week.

Observed deviations were consistent with missing or unreported information rather than implausible records.

3.1.3 Data Profiling

Data profiling was conducted to obtain a concise structural and statistical overview of the dataset prior to cleaning. This phase complements the data quality assessment by providing descriptive evidence on attribute behavior and distributions.

Structural Overview The dataset contains 319,073 records and 17 attributes, including identifiers, offense-related fields, temporal components, and spatial coordinates. Numerical and categorical data types are consistently represented, with no structural inconsistencies detected.

Single-Column Analysis Cardinality, distinctness, and uniqueness metrics were computed for each attribute. Temporal variables show low cardinality as expected, spatial attributes exhibit moderate distinctness due to repeated locations, and textual fields display high cardinality with low uniqueness.

Distributional and Geospatial Analysis Descriptive statistics confirmed bounded and interpretable temporal ranges. Spatial coordinates cluster within a narrow geographic window consistent with Boston, with sparse regions largely explained by missing values rather than outliers.

Automated Profiling Automated profiling using `ydata-profiling` corroborated manual findings, highlighting missing-value concentrations, repeated identifiers, and minor textual inconsistencies. These results informed the targeted cleaning and record linkage strategies applied in subsequent stages.

3.2 Data Cleaning

Data cleaning was performed to address the systematic issues identified during data profiling and quality assessment. The objective of this stage is to correct, standardize, and rationalize the dataset while preserving its original semantics and analytical value.

Cleaning operations were designed to be rule-based, reproducible, and minimally invasive, focusing on missing values, placeholder encodings, textual inconsistencies, and structural anomalies revealed by automated and manual checks.

Handling Missing Values Missing values were concentrated in a small subset of variables, particularly `SHOOTING`, spatial attributes, and address-related fields. Given the domain meaning of the `SHOOTING` variable, missing values were interpreted as non-shooting incidents and explicitly encoded as `'N'`. Other missing values were retained as `NaN` to avoid introducing artificial information.

Placeholder and Invalid Encodings Several fields contained placeholder values used to represent missing information, including invalid coordinate pairs such as `(0,0)` or `(-1,-1)`. These values were systematically detected and converted to `NaN` to prevent downstream spatial inconsistencies. Textual placeholders such as `'nan'`, `'none'`, and empty strings were similarly normalized.

Duplicate Records Exact duplicate rows were rare and removed when detected. However, repeated `INCIDENT_NUMBER` values were preserved, as they reflect legitimate multi-record representations of a single incident rather than data duplication errors.

Reference Integrity Checks Offense codes were validated against the official reference table to ensure referential integrity. All offense codes present in the crime dataset were confirmed to exist in the reference file, eliminating the need for record removal at this stage.

Overall, data cleaning focused on correcting clearly invalid representations while retaining all analytically meaningful records for subsequent transformation and modeling.

3.2.1 Data Transformation and Standardization

Following cleaning, a series of transformations were applied to improve consistency, interpretability, and analytical usability. These transformations standardize formats, harmonize categorical values, and enrich the dataset with derived attributes.

Schema Harmonization Column names were standardized to a consistent naming convention to improve clarity and downstream compatibility. Temporal fields were explicitly converted to appropriate datetime formats, and numeric columns were coerced into numeric types where applicable.

Text Normalization Categorical text fields such as `STREET`, `DISTRICT`, and offense descriptions were normalized using rule-based transformations. These included lowercasing, removal of punctuation, standardization of common abbreviations, and normalization of directional indicators. This step reduced superficial variation while preserving semantic meaning.

Typo Detection and Correction Rare textual variants were identified using fuzzy string similarity clustering. Within each cluster, infrequent variants were replaced with the most common normalized form. This approach corrected typographical errors and minor spelling variations without collapsing legitimately distinct categories.

Spatial Standardization Location strings were parsed to extract latitude and longitude values when explicit coordinates were missing. Extracted coordinates were used to fill missing spatial fields, and temporary helper variables were removed after validation to maintain dataset cleanliness.

Feature Enrichment A binary holiday indicator was derived from the occurrence timestamp using an external calendar reference. This additional feature supports temporal pattern analysis without altering the original temporal variables.

3.2.2 Error Detection and Correction

Error detection and correction were performed using a combination of rule-based logic, statistical validation, and model-assisted techniques. Each correction strategy was chosen based on the structure of the error and its observed relationship with other variables.

Missing Values Missing values were analyzed separately for categorical, numerical, temporal, and spatial attributes. For the `SHOOTING` variable, missing entries were interpreted as non-shooting incidents and explicitly encoded as ‘‘N’’, based on domain semantics and empirical validation. This decision significantly improved completeness without introducing ambiguity.

For administrative and location-related fields (`DISTRICT`, `REPORTING_AREA`, `STREET`), missing values were imputed using hierarchical and locality-aware strategies. When spatial coordinates were available, the most frequent categorical values within the same geographic neighborhood were used. This approach preserves local structure and avoids unrealistic global imputation.

Categorical Imputation Categorical variables with structured dependencies, such as `UCR_PART`, were imputed using supervised classification. A multi-class model was trained using offense-related attributes (`OFFENSE_CODE`, `OFFENSE_DESCRIPTION`, `Crime.Type`), ensuring that imputations were consistent with observed offense semantics.

This strategy outperformed naive frequency-based filling while maintaining interpretability and alignment with official crime categorizations.

Numerical and Spatial Imputation Missing numerical values, particularly latitude and longitude, were imputed using a combination of group-level statistics and multivariate modeling. Median coordinates were computed at multiple aggregation levels (street, reporting area, district) and applied hierarchically. For remaining gaps, multivariate imputation via chained equations (MICE) was used to capture dependencies between spatial and contextual variables.

Outlier Detection Outliers in spatial coordinates and temporal attributes were detected using both rule-based thresholds and unsupervised models. Geographic bounds were enforced based on known city limits. Isolation Forest was applied to latitude-longitude pairs to identify anomalous spatial patterns. Detected outliers were removed only when confirmed by multiple criteria, avoiding over-pruning of valid but rare observations.

Semantic Inconsistencies Semantic inconsistencies between `OFFENSE_CODE`, `OFFENSE_DESCRIPTION`, and `OFFENSE_CODE_GROUP` were systematically identified. Rather than forcing normalization, these inconsistencies were documented and resolved through reference alignment and fuzzy matching, preserving legitimate alternative descriptions tied to the same offense code.

3.2.3 Data Deduplication

Data deduplication was performed using a staged strategy designed to remove true duplicates while avoiding the erroneous deletion of valid records. This approach reflects the fact that a single real-world incident may legitimately generate multiple records in the dataset.

Exact Duplicate Removal In a first step, exact duplicate rows were identified and removed deterministically. Only records that were fully identical across all attributes were eliminated. This operation removes redundant copies without affecting the semantic content of the dataset.

Rule-Based Deduplication The `INCIDENT_NUMBER` field was not assumed to be globally unique. Instead, it was treated as an event identifier that may correspond to multiple records. To address inconsistencies, rule-based checks were applied to identify cases where the same incident identifier appeared with conflicting temporal or spatial information.

Records were flagged when the same identifier was associated with multiple timestamps or distinct locations. In such cases, only records with consistent time and place information were retained, while inconsistent duplicates were removed. This step enforces logical coherence without collapsing valid multi-record incidents.

Non-Exact Deduplication via Record Linkage To detect near-duplicate records not captured by exact or rule-based methods, a probabilistic record linkage approach was employed. Candidate record pairs were generated using blocking on temporal, spatial, and administrative attributes to reduce computational complexity.

Similarity features were computed across latitude, longitude, hour, street name, and offense code. A supervised Logistic Regression model was trained to classify record pairs as duplicates or non-duplicates. Highly similar pairs were clustered using a graph-based approach, where each connected component represents a group of duplicate records.

For each cluster, a single representative record was retained, selected as the earliest event in time. Records not belonging to any cluster were kept unchanged. This strategy allows robust duplicate resolution under uncertainty while preserving valid observations.

Post-Deduplication Assessment After completing all deduplication stages, no duplicate records remained in the dataset. The procedure successfully reduced redundancy while maintaining consistency with the underlying event structure, providing a clean and reliable basis for downstream analysis.

4 Results

4.1 Data Analysis Setup

A supervised classification task was selected to evaluate the impact of data cleaning on downstream analytical performance. The objective is to predict a discrete crime category and to assess whether improvements in data quality lead to more stable and consistent class separation.

The target variable is **Crime.Type**, a categorical attribute derived from offense codes. Due to the presence of highly underrepresented classes, rare categories were grouped into an **Other** class. This choice enables a stable stratified train-test split and reduces noise caused by extremely sparse labels.

The feature set includes:

- Temporal features: **Year**, **Month**, **Hour**
- Spatial and administrative features: **District**, **Reporting Area**
- Contextual feature: **Shooting**

The same analysis pipeline was applied to both the dirty and cleaned datasets to ensure a controlled and fair comparison. Numerical features were imputed using the median, while categorical features were imputed using the most frequent value. These choices were made to ensure robustness to outliers and to avoid introducing additional modeling assumptions at the analysis stage.

Categorical variables were one-hot encoded, and numerical features were standardized to ensure comparability and compatibility with linear models.

A Logistic Regression classifier with class balancing was selected as the primary model. This model was chosen because it is interpretable, computationally efficient, and sensitive to data quality variations, making it suitable for isolating the effect of data cleaning independently of model complexity.

Hyperparameters were selected using cross-validation. Model performance was evaluated exclusively on a held-out test set to ensure unbiased comparison between the dirty and cleaned datasets.

4.2 Results Comparison

Model performance was evaluated using the macro F1-score, which is appropriate in the presence of strong class imbalance, as it assigns equal importance to all classes regardless of their frequency.

Dataset	Macro F1-score
Dirty	0.0079
Cleaned	0.0131

Table 1: Macro F1-score comparison between dirty and cleaned datasets

The cleaned dataset shows an absolute improvement of +0.0052 in macro F1-score compared to the dirty dataset. Although the absolute values remain low, the relative improvement is consistent and systematic.

4.3 Discussion

The results presented above provide empirical evidence for the effectiveness of the proposed data quality pipeline. The absolute performance values are low, which is expected given the characteristics of the dataset. The data is not specifically designed for predictive modeling, the target variable is highly imbalanced, and only a limited set of explanatory features was used.

Nevertheless, the cleaned dataset consistently outperforms the dirty one. This confirms that improvements in data quality translate into measurable benefits in downstream analytical tasks, even when predictive performance remains limited.

5 Conclusion

This project implements a complete, justified, and reproducible data quality pipeline. The data cleaning process was transparent, explainable, and traceable. Its effectiveness was validated through a controlled comparison using a classification task.

The primary objective of the project—the implementation of a correct, well-documented, and explainable data preparation pipeline—was achieved.